

MANIPAL UNIVERSITY JAIPUR

Department of Data Science Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech CSE - DS
Course Name:	Statistical Methods and Probability Theory
Course Code:	MAS2001
LTPC (Lecture Tutorial Practical Credits):	2 0 3
Session:	Jan-May 2025
Class:	IV Semester
Course Coordinator:	
Course Instructor(s):	Dr. Reema Jain
Additional Practitioner(s) – if any (Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):	To be identified and appointed later

- B. Introduction:** This course is offered by the Dept. of Mathematics & Statistics as a regular course to make the students acquainted with the subject of probability and statistics at an early stage. Probability and statistics are important foundations for computer science fields such as machine learning, artificial intelligence, computer graphics, randomized algorithms, image processing, and scientific simulations. In this course, students will expand their knowledge of probabilistic methods and apply them to diverse computational problems. The first part of the course offers in depth knowledge of probability theory (random event, probability, characteristics of random variables, probability distributions and moment generating functions) which is necessary for simulation of random processes. In the second part, sampling theory is discussed. Each concept is explained through various examples and application-oriented problems.

- C. Course Outcomes:** At the end of the course, students will be able to:

CO Statement	CO	Level	Target Attainment %	Target Attainment level
Describe the concepts of probability and random variables to analyze and solve a wide range of real-world problems under uncertainty principles and make decisions in practical scenarios.	2001.1	L2 (Understand)	At least 85%	3
Employ probability distributions to effectively model and analyze real-world engineering problems.	2001.2	L3 (Apply)	At least 70%	2
Describe point and interval estimators for parameter estimation to provide reliable and	2001.3	L4 (Analysis)	At least 70%	2

accurate conclusions in industry and business for effective decision-making.				
Formulate and test the Hypothesis related to engineering problems and enhance decision-making and employability.	2001.4	L5 (Evaluate)	At least 65%	I

D. PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

(B.Tech in CSE – DSE): At the end of the programme, students will be able to:

- [PO. 1] Engineering knowledge:** apply the knowledge of mathematics, computer science, and communication engineering fundamentals to the solution of complex engineering problems.
- [PO. 2] Problem analysis:** identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using basic principles of mathematics, computing techniques and communication engineering principles.
- [PO. 3] Design/development of solutions:** design solutions for complex engineering problems and design system components or processes that meet the specified requirements with appropriate consideration for law, safety, cultural & societal obligations with environmental considerations.
- [PO. 4] Conduct investigations of complex problems:** use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- [PO. 5] Modern tool usage:** create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- [PO. 6] The engineer and society:** apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- [PO. 7] Environment and sustainability:** understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- [PO. 8] Ethics:** apply ethical principles and commit to professional ethics, responsibilities and norms of the engineering practice.
- [PO. 9] Individual and teamwork:** function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- [PO. 10] Communication:** communicate effectively for all engineering processes & activities with the peer engineering team, community and with society at large. Clarity of thoughts, being able to comprehend and formulate effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- [PO. 11] Project management and finance:** demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in varied environments.
- [PO. 12] Life-long learning:** recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

- [PSO.1].** Understand the role of mathematics, statistics, and AI techniques in the field of data science & engineering.
- [PSO.2].** Apply the acquired knowledge and expertise to perform data analytics tasks for multidimensional data sets.
- [PSO.3].** Develop effective and scalable industrial solutions for real-world socio-economic problems using data analytics tools and techniques.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): Quiz, MOOC, Research work Assignment, Video Assignment, etc.	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. SYLLABUS:

Probability Theory and Random Variables: Probability (Only One Lecture), Random variables, Cumulative distribution functions, Discrete random variables, Continuous random variables, Independent random variables, Probability mass and density functions, Expectation of random variables, Chebyshev's inequality, Central limit theorem. Probability distribution: Binomial, Poisson, Uniform, Normal, Exponential Theory of Estimation: Maximum Likelihood and method moment estimation, Sufficient statistics, Bayesian estimation, Confidence intervals for means. Tests of Statistical Hypothesis: Introduction, Parameter and Statistic, Standard error, Statistical hypotheses, Critical region, Tests of hypotheses and significance, Type I and Type II errors, level of significance. Test about one mean, Test about equality of two means, Test of variances, Chi square test, Analysis of Variance.

G. References:

1. Goon, A.M., Gupta, M.K. and Dasgupta, B., An Outline of Statistical Theory, Vol. II, 3rd Edn. World Press, Kolkata, 2005.
2. Kendall and Stuart, Advanced Theory of Statistics Vol.-II, Charles Griffin & Co. Ltd, London, 1961.
3. Casella, G. and Berger, R.L., Statistical Inference, Second Edn. Thomson Duxbury, 2002.
4. Hogg, R.V. and Tanis, E.A., Probability and Statistical Inference, 3rd Edn. Macmillan Publishing Co. Inc., 1988.
5. Feller, W., An Introduction to Probability Theory and Its Applications, Vol. 1, 3rd Edition, John Wiley, 1968.

H. Lecture Plan:

Lect. No.	Topics	Session Outcome	CO	Mode of delivery	Mode of Assessing CO
1	Introduction of the Course	Develop the understanding about the course	MAS2001.1	Lecture	NA
2	Probability Theory and Random Variables: Probability (Only One Lecture)	Revise the concept of probability theory	MAS2001.1	Lecture	MTE, CWS, ETE
3-4	Random variables, Cumulative distribution functions, Probability mass and density functions	Learn about Random Variable	MAS2001.1	Lecture	MTE, CWS, ETE
5-6	Discrete random variables, Continuous random variables,	Understand the types of random variables	MAS2001.1	Lecture	MTE, CWS, ETE
5	Expectation of random variables	Learn about properties of random variables	MAS2001.1	Lecture	MTE, CWS, ETE
6	Expectation of random variables, Independent random variables	Learn about properties of random variables	MAS2001.1	Lecture	MTE, CWS, ETE
7	Chebyshev's inequality	Learn about properties of random variables	MAS2001.1	Lecture	MTE, CWS, ETE
8	Tutorial-Problem Solving	Learn about properties of random variables	MAS2001.1	Lecture/ Flipped Classroom	MTE, CWS, ETE
9	Binomial distribution	Acquire the knowledge about distributions of data	MAS2001.2	Lecture	MTE, CWS, ETE
10	Poisson distribution	Acquire the knowledge about distributions of data	MAS2001.2	Lecture, Peer teaching	MTE, CWS, ETE

11	Uniform distribution	Acquire the knowledge about distributions of data	MAS2001.2	Lecture/ Flipped Classroom	MTE, CWS, ETE
12	Normal distribution	Acquire the knowledge about distributions of data	MAS2001.2	Lecture	MTE, CWS, ETE
13	Exponential distribution	Acquire the knowledge about distributions of data	MAS2001.2	Lecture	MTE, CWS, ETE
14	Tutorial-Problem Solving	Acquire the knowledge about distributions of data	MAS2001.2	Lecture	MTE, CWS, ETE
15	Theory of Estimation: Introduction, Parameter and Statistic	Gain knowledge about estimation theory	MAS2001.3	Lecture	MTE, CWS, ETE
16	Central limit theorem	Acquire the knowledge about distributions of data	MAS2001.1	Peer teaching	MTE, CWS, ETE
17	Sufficient statistics	Gain knowledge about estimation theory	MAS2001.3	Lecture	MTE, CWS, ETE
18-19	Maximum Likelihood	Gain knowledge about estimation theory	MAS2001.3	Lecture	MTE, CWS, ETE
20	Method Moment Estimation	Gain knowledge about estimation theory	MAS2001.3	Group-teaching and learning	MTE, CWS, ETE
21	Bayesian estimation	Gain knowledge about Bayesian estimation theory	MAS2001.3	Lecture	MTE, CWS, ETE
22	Tutorial-Problem Solving	Gain knowledge about estimation theory	MAS2001.3	Lecture	MTE, CWS, ETE
23	Confidence intervals for means	Develop the notion of interval estimator for unknown parameters	MAS2001.3	Peer teaching	MTE, CWS, ETE
24	Confidence intervals for means	Develop the notion of interval estimator for unknown parameters	MAS2001.3	Lecture	MTE, CWS, ETE

25	Confidence intervals for means	Develop the notion of interval estimator for unknown parameters	MAS2001.4	Lecture	MTE, CWS, ETE
26	Tests of Statistical Hypothesis: Introduction	Elaborate the concept of testing of hypothesis	MAS2001.4	Lecture/ Flipped Classroom	MTE, CWS, ETE
27	Standard error, Statistical hypotheses, Critical region, level of significance	Elaborate the concept of testing of hypothesis	MAS2001.4	Lecture	MTE, CWS, ETE
28	Tutorial-Problem Solving	Develop the notion of interval estimators for unknown parameters	MAS2001.4	Learning through problem-solving	CWS, ETE
29	Type I and Type II errors	Elaborate the concept of testing of hypothesis	MAS2001.4	Lecture	CWS, ETE
30	Test about one mean	Gain knowledge about implementation of t-test	MAS2001.4	Group-teaching and learning	CWS, ETE
31	Test about equality of two means	Gain knowledge about implementation of t-test	MAS2001.4	Lecture	CWS, ETE
32	Test about equality of two means	Gain knowledge about implementation of t-test	MAS2001.4	Lecture	CWS, ETE
33	Test of variances	Gain knowledge about the implementation of the F-test	MAS2001.4	Peer teaching	CWS, ETE
34	Chi-square test	Gain knowledge about the implementation of the Chi-square test	MAS2001.4	Lecture	CWS, ETE
35	Chi-square test	Gain knowledge about the implementation of the Chi-square test	MAS2001.4	Lecture	CWS, ETE

36	Analysis of Variance (ANOVA)	Gain knowledge about implementation of ANOVA	MAS2001.4	Peer teaching	CWS, ETE
37	Analysis of Variance (ANOVA)	Gain knowledge about the implementation of ANOVA	MAS2001.4	Lecture	CWS, ETE
38	Analysis of Variance (ANOVA)	Gain knowledge about the implementation of ANOVA	MAS2001.4	Lecture	CWS, ETE
39	Tutorial-Problem Solving	Able to apply testing of hypothesis in decision-making	MAS2001.4	Lecture	CWS, ETE
End Term Examination					

I. Course Articulation Matrix: (Mapping of COs with POs and PSOs)

CO	Statement	Correlation with Program Outcomes and Program Specific Outcomes														
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
MAS2001.1	Apply the concepts of probability and random variables to analyze and solve a wide range of real-world problems under principles of uncertainty and make decisions in practical scenarios.	2	1	1			1		1				1			
MAS2001.2	Employ probability distributions to effectively model and analyze real-world engineering problems.	1	2	3	1	1	2		1	1		1	1	2		
MAS2001.3	Apply point and interval estimators for parameter estimation to provide reliable and accurate conclusions in industry	2	2	3	2	3	2		1	2	1		1	2	1	

	and business for effective decision making.															
MAS2001.4	Formulate and test the Hypothesis related to engineering problems and enhance the decision making and employability.	1	3	3	2	3	2	1	1	1				1	1	1

1: Low Correlation 2: Moderate Correlation 3: Substantial Correlation